

MAHAK SHARMA

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PROFESSIONAL SUMMARY

AI/ML researcher and Machine Learning Engineer with a research and industry background spanning Artificial Intelligence & Machine Learning, Computer Vision, Human-AI Collaboration, NLP, Continual Learning, and AI-Driven Cybersecurity. Author of two peer-reviewed papers accepted for publication in AI & Society and Computers & Security, with an additional manuscript in preparation. Brings research experience, industry experience in ML and AI systems, professional certifications in machine learning and cybersecurity, and recognition for both research excellence and professional performance.

RESEARCH INTERESTS

Artificial Intelligence & Machine Learning (Computer Vision, Deep Learning, Continual Learning) | Human-AI Collaboration | AI-Driven Cybersecurity

EDUCATION

SUNY University at Buffalo, USA

July 2023 – June 2024

MS in Management Information Systems

Relevant areas: IT & Cloud Infrastructure Management, Distributed Computing & Big Data, Database Management Systems, Systems Analysis & Design, Online Analytical Processing & Data Warehousing, Predictive Analytics

Medi-Caps University India, India

August 2019 – April 2022

Masters in Computer Application

Relevant areas: Advanced Programming, Data Structures, Design and Analysis of Algorithms, Mathematics of Computer Applications, Computer Organization and Architecture, Internet of Things, Data Warehousing and Mining, Computer Networks, Computer Oriented Numerical and Statistical Methods

RESEARCH EXPERIENCE

Medi-Caps University, India

Research Assistant — Computer Vision & Deep Learning

July 2021 – April 2022

- Built and optimized deep learning pipelines (CNN-BiLSTM with attention) for posed vs. spontaneous expression classification, achieving performance gains through systematic hyperparameter tuning and architecture modifications.
- Processed and labeled 5,000+ video clips from MMEW and CAS(ME)² datasets—extracting facial Action Units, optical flow, and head pose features using OpenFace for model training.
- Ran ablation experiments isolating temporal features from static features to understand what actually differentiates real from fake expressions.
- Owned the lab's training infrastructure—managed GPU servers, tracked experiments, and maintained all code and documentation for team-wide use.

Medi-Caps University, India

Research Assistant — Cybersecurity & Cloud Access Control

July 2020 – June 2021

- Implemented multi-factor authentication modules and integrated them with RBAC and ABAC policies on AWS/Azure test environments for comparative security analysis.
- Wrote Python scripts to generate simulated authentication requests, modeling burst traffic and concurrent user sessions to evaluate system throughput and failure rates.
- Measured authentication latency (p50/p95), false acceptance rate (FAR), and false rejection rate (FRR) across multiple hybrid framework configurations to identify optimal policy combinations.
- Maintained the complete experimental codebase, test infrastructure logs, and configuration matrices — enabling reproducible results for two research publications and technical reports.

PUBLICATIONS AND MANUSCRIPTS

- Sharma, M. “Detecting Trust Miscalibration in Human-AI Collaboration Using Multimodal Emotion Cues: A Machine Learning Framework.” AI & Society, Accepted, 2026.
- Sharma, M. “An AI-Augmented Governance Architecture for Dynamic Cybersecurity Risk Management in SMEs: Design, Implementation, and Longitudinal Assessment.” Computers & Security, Accepted, 2026.
- Sharma, M. “Replayless Structural Consolidation: Continual Learning via Dynamic Network Motifs.” *Manuscript in preparation.*

PROFESSIONAL EXPERIENCE

Amazon (via Forman Tech), USA

Machine Learning Engineer

July 2025 – Present

- Designed and evaluated ML-based recommendation and ranking systems using customer behavior, purchase history, and product metadata to improve personalization.
- Developed NLP models for product search and query understanding, including query classification, semantic search, and product ranking, with relevance-focused experimentation.
- Built scalable Python/SQL/PySpark data pipelines and trained deep learning models on AWS SageMaker, applying hyperparameter tuning, automated monitoring, and drift detection.

Community Dreams Foundation, USA

Program Manager

October 2024 – July 2025

- Managed AI/ML-focused technology programs from requirements definition through implementation, coordinating technical deliverables, timelines, and project priorities.
- Worked with engineers, data scientists, and community stakeholders to translate program requirements into scalable technology solutions and implementation plans.
- Defined and monitored program KPIs and performance metrics, using data-driven analysis to evaluate outcomes, identify gaps, and guide project improvements.

Tata Consultancy Services (TCS), INDIA

Systems Engineer

July 2022 – June 2023

- Developed Python- and SQL-based machine learning workflows for data preprocessing, feature engineering, and predictive modeling across structured enterprise datasets.
- Built and optimized ETL/data pipelines to transform, validate, and prepare large-scale datasets for machine learning and analytical workloads.
- Evaluated model performance using accuracy and error metrics, conducting feature-level analysis and optimization to improve predictive results and support production analytics.

RESEARCH AND METHODOLOGICAL SKILLS

- **Research:** Research Design, Experimental Design, Literature Review, Systematic Literature Review, Ablation Studies, Comparative Analysis
- **Analysis:** Statistical Analysis, Model Evaluation, Feature Analysis, Error Analysis, Benchmarking
- **Development:** Framework & Architecture Development, Algorithm Development, Reproducible Research
- **Communication:** Academic Writing, Technical Writing, Research Presentations, Research Dissemination

TECHNICAL SKILLS

- **AI & Machine Learning:** Machine Learning, Deep Learning, Neural Networks, Continual Learning, Recommender Systems, Ranking & Personalization, Anomaly Detection, Feature Engineering.
- **Computer Vision & Multimodal AI:** Facial Expression Recognition, Video Analysis, Facial Action Units, Optical Flow, Head Pose Estimation, CNNs, CNN-BiLSTM, OpenFace
- **Natural Language Processing:** NLP, Query Classification, Semantic Search, Query Understanding, Text Classification, Product Ranking
- **Cybersecurity & AI Security:** Cloud Security, Multi-Factor Authentication, RBAC, ABAC, Access Control, Cybersecurity Risk Management, AI Governance
- **ML Engineering & MLOps:** TensorFlow, Scikit-learn, Model Training, Model Monitoring, Drift Detection, Automated Training, Experiment Tracking

CERTIFICATES

- AWS Certified Machine Learning Engineer – Associate (MLA-C01) — Amazon Web Services (AWS)
- CompTIA Security+ — CompTIA
- Certified Information Systems Auditor (CISA) — ISACA

AWARDS AND ACHIEVEMENTS

- Best Research Presentation Award — Oral Category, Medi-Caps University
- TCS GEMS Award — Employee Recognition